

TOTHBOT — KRAKEN SYSTEM ARCHITECTURE OVERVIEW

The Complete System — All Architectural Decisions + All Flow Diagrams

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PURPOSE: This document captures all architectural decisions, FP/DP analyses, and system specifications from the TothBot Kraken architecture sessions of March 2026. Includes all six system flow diagrams showing connections between Kraken, TothBot, and CIATS as one unified organism.

Diagram	Title	Version Note
1	Kraken API Connection Architecture	dv1_7: A-10 executions seq gap + GetOpenOrders gap recovery
2	TothBot Trading Pipeline	dv1_6: per-pair instrument status check before Gate 1
3	Exit Architecture: Ground Zero Redesign	dv1_6: title updated
4	Regime Taxonomy: 6-Regime Grid + Two-Level Tagging	dv1_6: title updated
5	CIATS Architecture: Data Flows and Improvement Cycle	dv1_6: title updated
6	Complete System: One CIATS-Owned Organism	dv1_7: A-10 integrated

1. System Identity and Mission

1.1 What TothBot Is

TothBot is an automated cryptocurrency spot trading system on Kraken exchange. It runs 24/7/365 on a Hetzner VPS, processes 5-minute OHLC candle close events as its system clock, and executes long-only trades using a momentum-based signal engine.

TothBot is not an external system connected to Kraken — it IS Kraken's event processor. Every action is a response to a Kraken push event. No timers. No polling. Idle and silent until Kraken speaks.

1.2 The One Organism Principle

Three functional zones, one organism:

Kraken	Generates events, executes orders, holds resting TP limit orders and off-book emergency SL stop orders on matching engine
TothBot	Processes Kraken events, executes trading strategy, manages positions
CIATS	Learns from event records, continuously improves processing rules

PRIMARY MISSION: Maximize profit. Minimize loss. With an emphasis on minimizing loss. Loss prevention takes priority over profit generation. Everything else is secondary.

1.3 Operating Environment

Exchange	Kraken Spot (Long Module — Short deferred)
Language	Python 3.12.3
VPS	Hetzner CPX22 — 87.99.141.44 — Ashburn VA
System Clock	Kraken OHLC 5-minute candle close event (WS v2 push)
Fee Baseline	Maker 0.16% / Taker 0.26% (CIATS monitors actuals)
Operation	24/7/365 — CIATS may add stand-down periods via PDCA
Starting Portfolio	\$1,000 Long Module
Sacred Constraint	Net 1:1.5 R:R — HARDCODED — only number not owned by CIATS

2. Kraken API Connection Architecture

All connection points verified across 3 independent analysis passes (15 sources) plus 2 deep validation sessions (144+ sources total) plus two 10-search FP/DP Kraken validation sessions [dv1_6, dv1_7] (239+ sources total).

2.1 Key Connection Points

Channel / Method	Direction	Purpose
PUBLIC WS v2		
instrument	INBOUND	Asset specs + per-pair status cache ★ dv1_6
status	INBOUND	Global engine state (online/post_only/cancel_only/maintenance)
ohlcv(5m)	INBOUND	SYSTEM CLOCK — pipeline trigger, fires on trade events
ohlcv(60)	INBOUND	1H HTF cache for Gate 4 ★ NEW dv1_5
ticker	INBOUND	Price data, event_trigger configurable (trades or bbo)
PRIVATE WS v2		
executions	INBOUND	Fills/expiration/SL-TP triggers, ratecounter:true (A-1), snap_orders:true (A-3), sequence gap detection
balances	INBOUND	Real-time settlement, sequence gap detection ★ A-9 dv1_6
add_order	OUTBOUND	Entry: limit/post_only/GTD/cl_ord_id/stp_type:cancel-newest
amend_order	OUTBOUND	PRIMARY amendment method (edit_order DEPRECATED ★ dv1_6)
batch_add	OUTBOUND	SL+TP atomic on fill
cancel_order	OUTBOUND	Exit leg cleanup, req_id tracked
batch_cancel	OUTBOUND	Full Halt multi-order cancel ★ dv1_6
REST		
GetWebSocketsToken	OUTBOUND	Startup/reconnect
GetOHLCData	OUTBOUND	Daily regime computation 00:00 UTC. CRITICAL: exclude response[-1]
GetOpenOrders	OUTBOUND	Restart reconcile fallback / executions gap recovery ★ dv1_7
GetAccountBalance	OUTBOUND	Restart reconcile fallback / balances gap recovery ★ dv1_6
CancelAllOrders	OUTBOUND	Kill switch emergency only
MATCHING ENGINE		
Resting TP + EmergSL	PASSIVE	Survive any TothBot failure independently — crash protection always active

2.2 Four Kraken Engine States

State	TothBot Behavior
online	All orders. Signal pipeline runs normally.
post_only	TothBot limit+post_only entries ARE valid. Trading continues.
cancel_only	No new entries. Exit Controller can cancel. Resting orders active.
maintenance	No orders, no cancels. Resting TP and emergSL active on engine.

2.3 API Rate Limits — Intermediate Tier (C-2)

Parameter	Intermediate Tier Value
Rate counter threshold (per pair)	125
Rate counter decay (per second)	2.34
Max open orders (per pair)	80
Add order fixed cost	+1
Amend order cost ★ dv1_6 (PRIMARY)	+1 fixed, +3(<5s), +2(<10s), +1(<15s) — 43% cheaper than edit_order for <5s orders

Edit order (DEPRECATED ★ dv1_6)	+1 fixed, +6(<5s), +5(<10s) — DO NOT USE
Batch add fixed cost	+(n/2) where n = orders in batch
Cancel order (< 5s old)	+8
Cancel order (< 10s old)	+6
Cancel order (< 300s old)	+1
Batch cancel cost ★ dv1_6	Cumulative sum of individual order cancel costs

2.4 WS Operational Requirements (A-1 through A-10)

A-1	Rate counter monitoring via executions channel executions subscription accepts ratecounter:true. Real-time rate counter per pair included at zero additional cost. CIATS EWMA Monitor adds rate_counter_by_pair as third monitoring signal. Assign to 0511003, 0211001.
A-2	req_id and cl_ord_id on all outbound WS messages All outbound messages carry req_id. All add_order/batch_add carry cl_ord_id (up to 18 chars). TothBot assigns cl_ord_id at dispatch before Kraken response. Position Mirror records cl_ord_id immediately — no async gap. WS Manager maintains pending-request registry. Assign to 0511001, 0511003, 0511005.
A-3	Round-trip latency monitoring and WS reconciliation All WS v2 responses include time_in and time_out. Logger records these. CIATS EWMA adds ws_roundtrip_latency_ms. executions channel supports snap_orders:true — preferred over REST GetOpenOrders for restart reconciliation. Assign to 0511001, 0511006, 0211001.
A-4	stp_type:cancel-newest on all add_order calls All add_order and batch_add include stp_type:cancel-newest. Self Trade Prevention prevents TothBot from filling against its own resting orders. Zero-cost prevention. Assign to 0511003.
A-5	Reconnect behavior distinction (1) Random drop: reconnect instantly, up to a handful of attempts. (2) After confirmed trading engine maintenance: minimum 5-second delay. Both implemented as separate paths in WS Manager. Assign to 0511001.
A-6 (dv1_5)	OTO conditional orders — REJECTED OTO uses limit price not actual fill price — violates sacred R:R. Creates Position Mirror race condition. batch_add on fill confirmed correct. See AR-008.
A-7 (dv1_6) ★	WS Ping / Keepalive Strategy Kraken WS server disconnects after ~60 seconds of inactivity. Cloudflare idle timeout ~100 seconds. WS Manager MUST send {"method":"ping"} on each active connection every 30 seconds. Expects {"method":"pong"} within 10 seconds. No pong = dead connection. Reconnect immediately. Application-level ping, distinct from TCP keepalive. Assign to 0511001, 1011002.
A-8 (dv1_6) ★	Zombie Connection Detection A WS connection can appear alive (ping/pong passing) while delivering no real market data. WS Manager MUST track last_real_data_time per connection using time.monotonic(). Only actual market data events reset this timestamp — heartbeat messages alone do NOT. If elapsed > 90 seconds: log ZOMBIE_CONNECTION_DETECTED, alert operator, reconnect. Assign to 0511001, 1011002.
A-9 (dv1_6) ★	Balances Channel Sequence Gap Detection Every balances channel message carries a sequence integer. WS Manager MUST track last-seen sequence. Any gap (skip > 1) indicates a missed ledger event. On gap: alert immediately, trigger REST GetAccountBalance reconciliation. Missed balance events cause incorrect available capital calculation — direct loss risk. Assign to 0511001, 1011002.
A-10 (dv1_7) ★	Executions Channel Sequence Gap Detection Every executions channel message carries a sequence integer. WS Manager MUST track last-seen sequence independently of balances. Any gap (skip > 1) indicates missed order/fill events. On gap: alert immediately, trigger REST GetOpenOrders reconciliation to restore Position Mirror accuracy. Missed execution events mean TothBot does not know a position is open or closed — direct loss risk. Higher severity than A-9: a missed fill leaves Position Mirror in stale state; the next pipeline evaluation operates on incorrect position data. Sequence tracking resets on each new subscription (reconnect). Do not gap-detect across sessions. Assign to 0511001, 1011002.

Diagram 1 — Kraken API Connection Architecture (dv1_7)

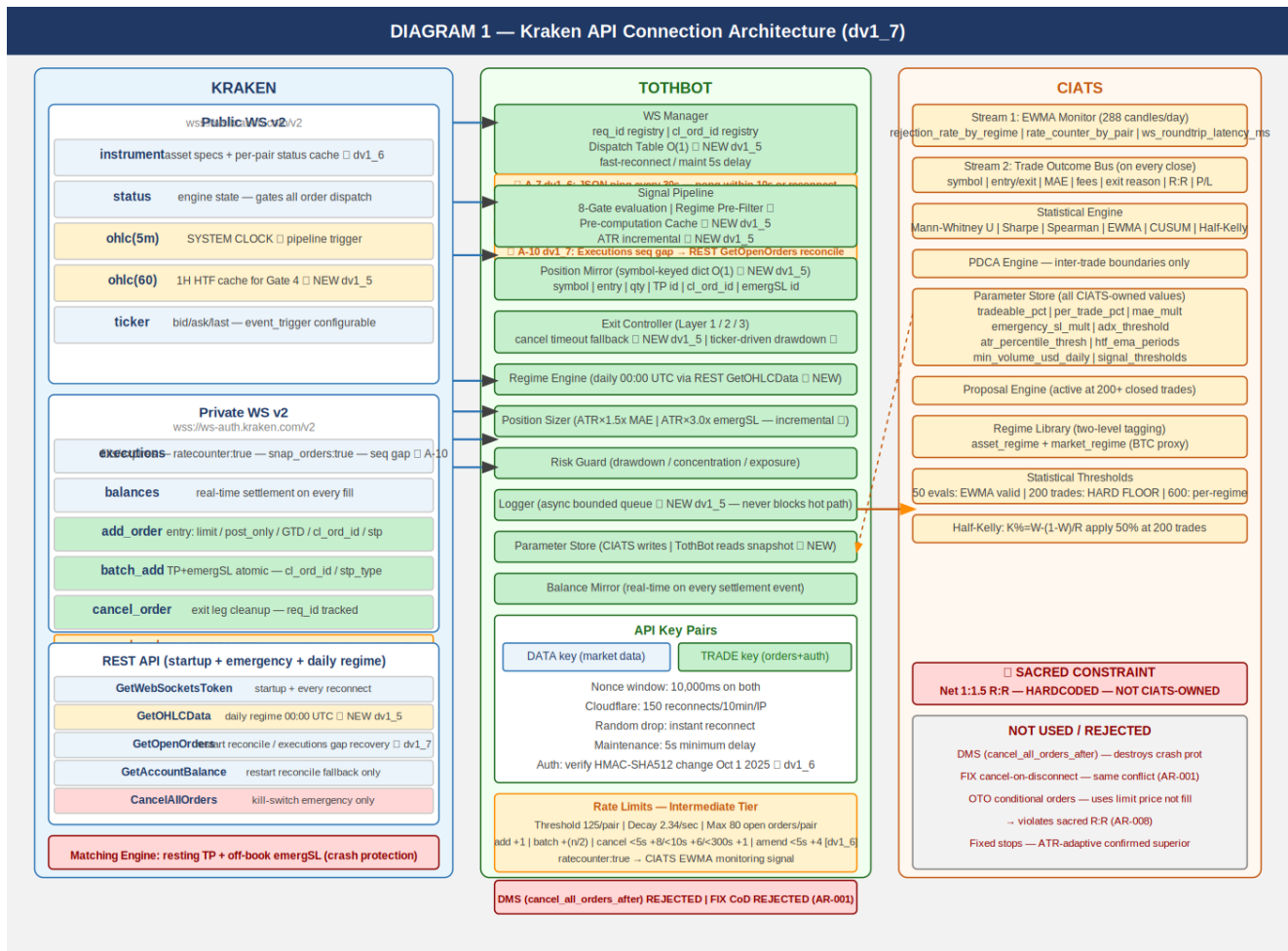


Diagram 1: Complete Kraken API connection map. A-10 (executions sequence gap detection) new in dv1_7.

2.5 Data Architecture: Pre-Computation Cache (NEW dv1_5 — I-2)

HOT PATH RULE: The 5-minute pipeline hot path reads cached values ONLY. No computation occurs during pipeline evaluation. All indicators maintained as incremental running calculations updated on each event trigger.

Value	Pre-computed by	Update Trigger	Cache Read On
Regime classification	Regime Engine	Daily 00:00 UTC REST GetOHLCData [CRITICAL]	Every 5m eval (Gate 3+5) ★
1H EMA(20) and EMA(50)	Signal Pipeline	1H candle close ohlc(60) WS event	Gate 4
ATR(14) per pair	Signal Pipeline	Each 5m candle close (incremental)	Gate 8 + Exit Controller
24h liquidity per pair (USD)	Signal Pipeline	Daily ticker or daily OHLC pull	Gate 2
Per-pair instrument status ★	WS Manager	instrument channel events (live)	Gates 1/2 pre-filter

3. TothBot Trading Pipeline

The pipeline fires on every 5-minute OHLC candle close event. Synchronous and event-driven. 8 gates. Gate 3 (Regime Pre-Filter) new in dv1_5. Per-pair status check new in dv1_6.

NOTE — OHLC fires on trade events, not wall clock. If no trades occur after an interval border, no message is sent until the next trade. TothBot must handle stalled system clock.

3.1 Gate Summary

Step	Name	Blocks / Acts When
PRE	Parameter Store Snapshot Frozen (NEW dv1_5 — I-1)	CIATS pairs captured as read-only snapshot at pipeline start. Consistent throughout full eval. CIATS wr
PRE	Pre-Computation Cache Read (NEW dv1_5 — I-2)	Hot path reads: regime cache 1H EMA cache ATR(14) running value 24h liquidity per-pair status. Zer
★ dv1_6	Per-pair Instrument Status Check	Skip pairs in reduce_only / work_in_progress / delisted / limit_only state. Only online or post_only pairs pro
Gate 1	State Machine	Kraken engine = maintenance or cancel_only → WAIT
Gate 2 ★	Liquidity Gate	Asset 24h volume < \$500k USD → SKIP: LIQUIDITY_REJECTED
Gate 3 ★ NEW	Regime Pre-Filter (NEW dv1_5 — I-1)	TRENDING_NEG (any vol) or NON_DIR+ELEVATED → STOP: REGIME_BLOCKED. Eliminates SSS co
Gate 4 ★	1H HTF Confirmation (was Gate 3)	1H EMA(20) <= EMA(50) for TRENDING_POS entries → SKIP: HTF_GATE_REJECTED. Cache read from
SSS	Signal Engine (SSS)	Scores 5-min candle. Only reached if Gate 3 pre-filter passed.
Gate 5	Selection Controller (was Gate 4)	Signal fails any of 6 quality gates → SKIP: SELECTION_REJECTED
Gate 6 ★	Regime Gate (was Gate 5)	NON_DIR+NORMAL: 50% position size. All NO ENTRY cases already blocked by Gate 3.
Gate 7	Risk Guard (was Gate 6)	Drawdown halt / concentration / exposure. Pass or HALT. ★ Ticker-driven continuous drawdown monitorin
Gate 8 ★	Position Sizer (was Gate 7)	MAE=ATR(14)×1.5x (incremental), emergSL=ATR(14)×3.0x, TP=R:R formula. Never blocks — only sizes.

Diagram 2 — TothBot Trading Pipeline

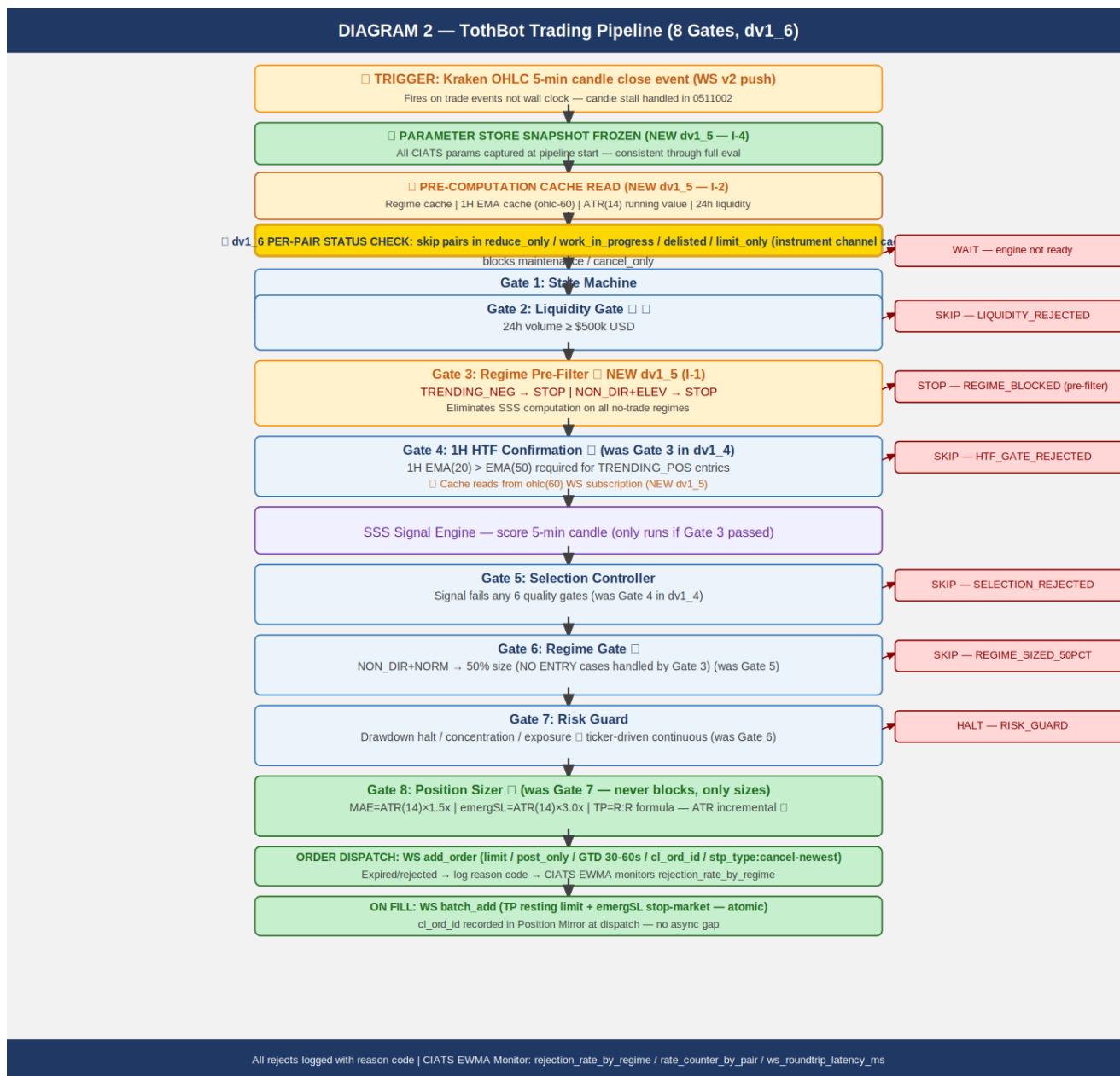


Diagram 2: 8-gate trading pipeline. Gate 3 Regime Pre-Filter and per-pair status check eliminate wasted computation.

4. Exit Architecture — Ground Zero Redesign for Kraken

Rebuilt from zero for Kraken. Exit Controller owns all normal exits. Kraken off-book emergency SL stop order is crash protection only. OTO evaluated and rejected (AR-008).

4.1 Three-Layer Summary

Layer	Name	Description
Layer 1a	Signal-Based Exit	Signal decay / time expiry / momentum loss. WS cancel TP + market sell.
Layer 1b	TP Fill	Resting TP limit on Kraken matching engine fills. EC cancels emergSL after fill confirm.
Layer 2 ★	MAE Threshold	Price - entry $\geq$ $ATR(14) \times 1.5x$. SEQUENCE CRITICAL: (1) cancel TP, (2) cancel emergSL, THEN (3) market sell.
Layer 3 ★	Emergency Brake	VPS/TothBot/internet failure ONLY. Off-book emergSL stop order at $ATR(14) \times 3.0x$. NEVER in normal operation.

4.2 R:R Formula

mae_pct	$= ATR(14) \times 1.5x / \text{entry_fill_price}$	CIATS-owned, uses ACTUAL fill price
net_loss	$= \text{mae_pct} + 0.16\% + 0.26\%$	Entry maker + MAE taker exit
net_gain	$= \text{net_loss} \times 1.5$	★★★ SACRED — HARDCODED ★★★
gross_target	$= \text{net_gain} + 0.16\%$	TP maker exit
emergSL_dist	$= ATR(14) \times 3.0x / \text{entry_fill_price}$	NEVER in R:R formula

Diagram 3 — Exit Architecture: Ground Zero Redesign

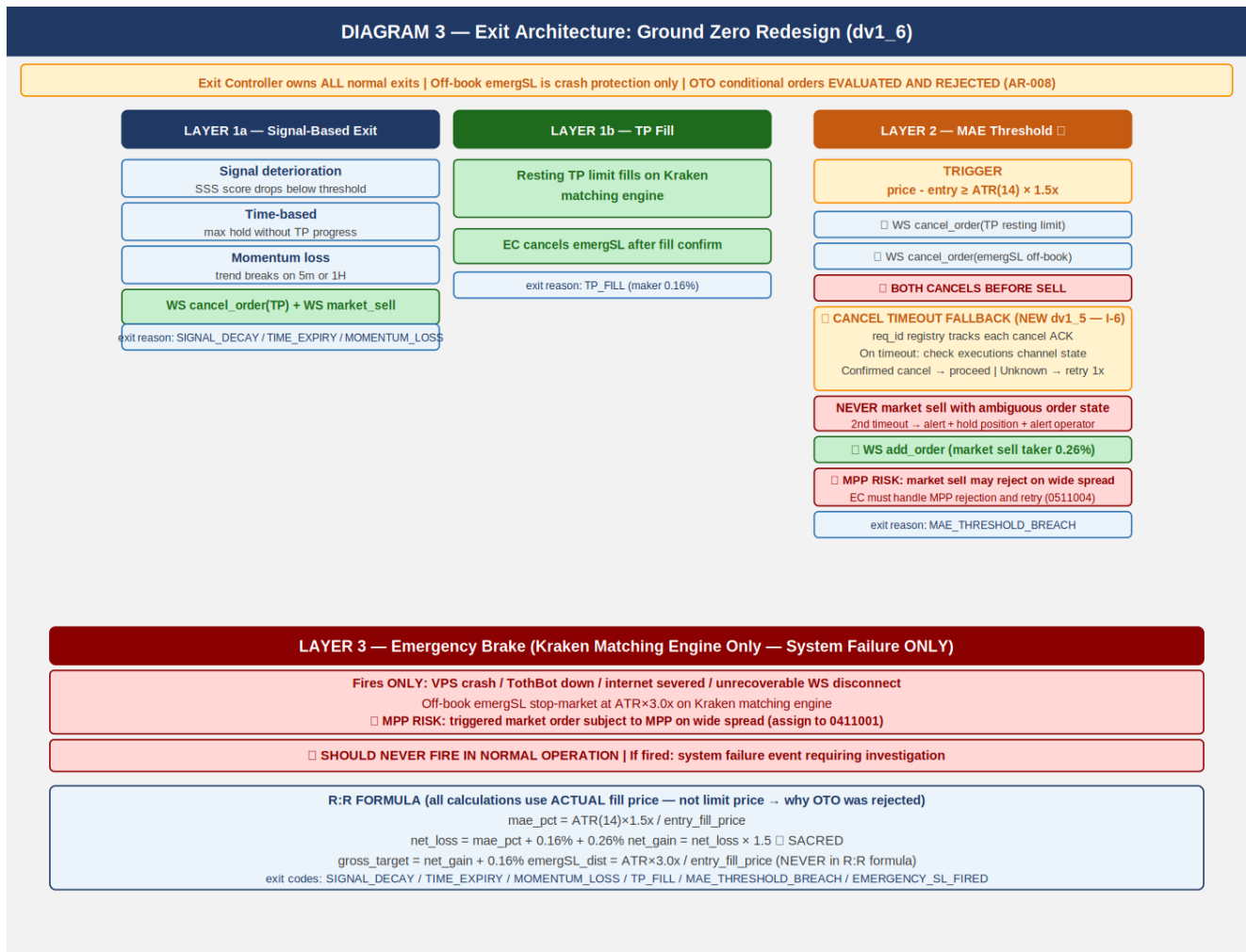


Diagram 3: Three-layer exit architecture. Exit Controller owns all normal exits. Layer 3 is crash protection only.

5. Regime Taxonomy

Six-regime grid is foundation of CIATS regime-segmented analysis. Every trade tagged with both asset_regime and market_regime (BTC proxy). Computed daily at 00:00 UTC.

CRITICAL dv1_6: The last entry (response[-1]) from REST GetOHLCData is ALWAYS the current not-yet-committed candle. MUST be excluded from ALL calculations (ADX, EMA, ATR). Only response[:-1] (committed candles) is valid for computation.

	NORMAL_VOL (ATR ≤ 67th)	ELEVATED_VOL (ATR > 67th)
TRENDING_POSITIVE (ADX>25, EMA20>EMA50)	FULL POSITION (Optimal) TRENDING_POS × NORMAL_VOL	FULL POSITION (ATR adapts) TRENDING_POS × ELEVATED_VOL
NON_DIRECTIONAL (ADX≤25)	50% POSITION NON_DIR × NORMAL_VOL	NO ENTRY NON_DIR × ELEVATED_VOL — Gate 3 blocks
TRENDING_NEGATIVE (ADX>25, EMA20<EMA50)	NO ENTRY TRENDING_NEG × NORMAL_VOL — Gate 3 blocks	NO ENTRY TRENDING_NEG × ELEVATED_VOL — Gate 3 blocks

All regime thresholds are CIATS-owned starting values: ADX threshold 25, ATR percentile 67th, EMA periods 20/50, rolling window 50 days. CIATS validates each after 600 closed trades (100+ per bucket).

Diagram 4 — Regime Taxonomy: 6-Regime Grid + Two-Level Tagging

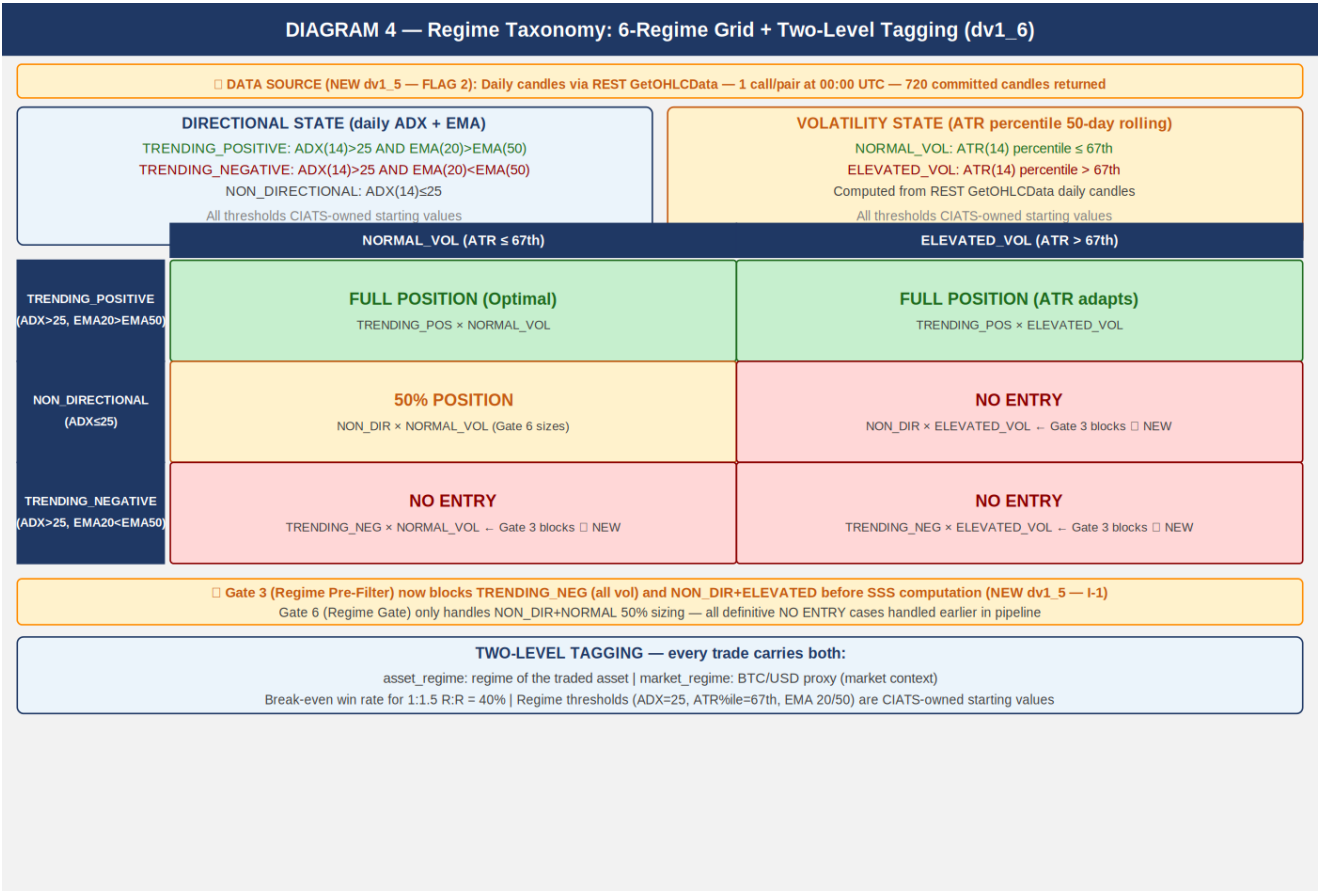


Diagram 4: 6-regime grid. Gate 3 pre-filter blocks TRENDING_NEG and NON_DIR+ELEVATED before SSS computation.

6. CIATS Architecture and Data Flows

CIATS is fully asynchronous. Never blocks the trading critical path. Reads only from the Logger. Writes only to the Parameter Store. Deming PDCA governs every change.

6.1 Two Data Streams

Stream	Description
Stream 1: EWMA Monitor	288 candle evals/day. Valid at 50 evals (~4 hrs). $\lambda=0.2$ [FP/DP validated dv1_6]. Three monitoring signals: (1) reject
Stream 2: Trade Outcome Buffer	Fires on each position close. Captures: symbol, entry/exit price, hold duration, MAE reached, fees, exit reason, both regime

6.2 Statistical Thresholds

Threshold	Meaning
50 candle evals	EWMA Monitor valid. Monitoring only. $\lambda=0.2$ validated.
200 trades HARD FLOOR	Inference valid. Proposal Engine activates. Half-Kelly activates. $SE(win_rate) \sim 3.5\%$, $SE(Sharpe) \sim 7.2\%$.
600 trades	Per-regime analysis. 100+ per bucket. MW power ~85% at $\alpha=0.01$.
Mann-Whitney alpha ★ dv1_6	$\alpha=0.01$ (one-sided). 99% confidence required for all parameter changes.
Spearman rho ★ dv1_6	$ \rho > 0.3$ AND $p < 0.05$ (both conditions required).
CUSUM parameters ★ dv1_6	$k=0.5 \cdot \sigma$, $h=4 \cdot \sigma$. Starting values. CIATS-owned.
PDCA minimum interval ★ dv1_6	Min 50 trades between any two parameter changes.

Diagram 5 — CIATS Architecture: Data Flows and Improvement Cycle

DIAGRAM 5 — CIATS Architecture: Data Flows and Improvement Cycle (dv1_6)

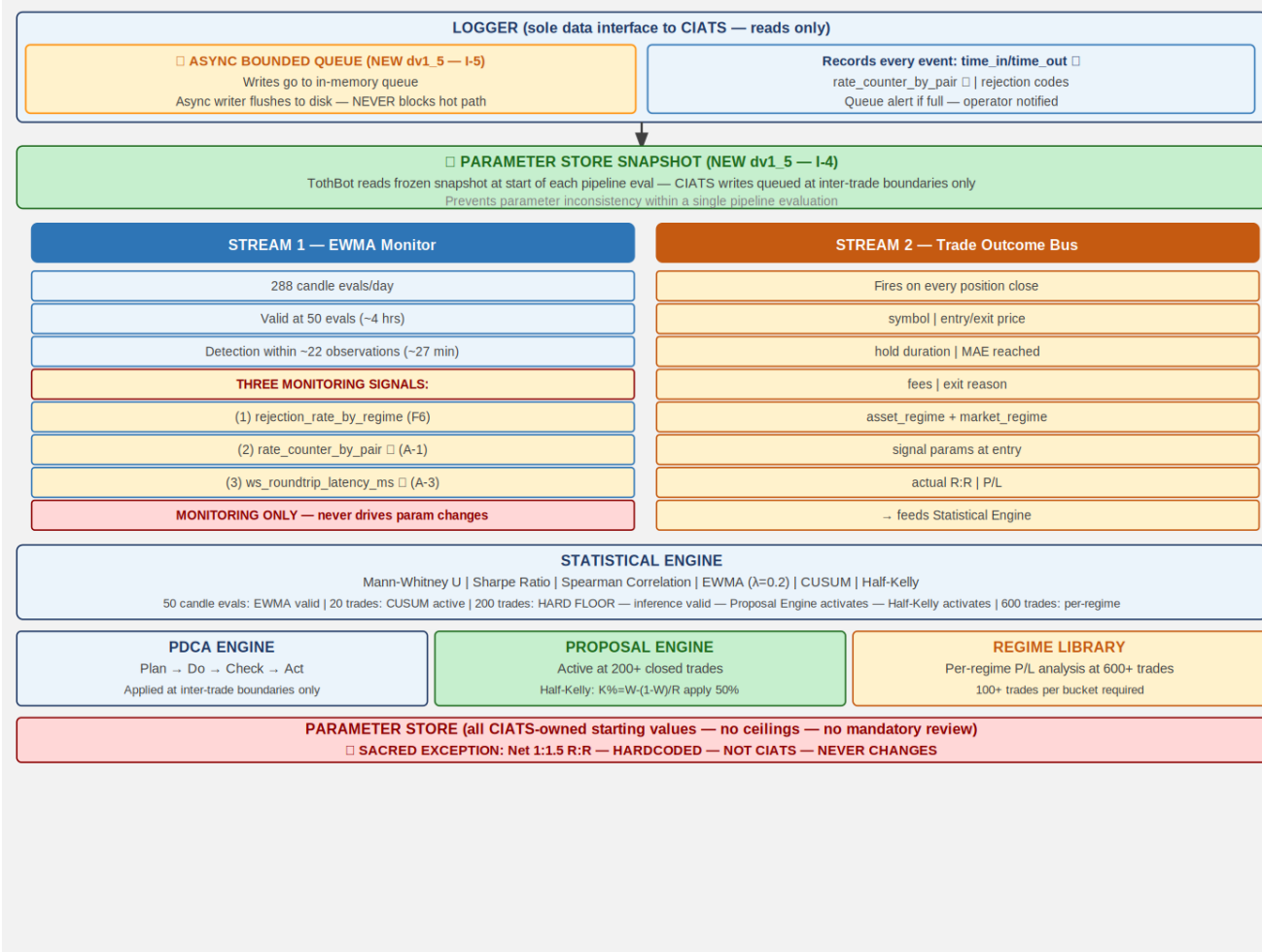


Diagram 5: CIATS architecture. Fully async. Reads Logger only. Writes Parameter Store only. PDCA governs every change.

7. Complete System: One CIATS-Owned Organism

CIATS OWNS EVERYTHING EXCEPT 1:1.5 R:R: All parameters, thresholds, sizing decisions, halt thresholds, multipliers, and trading rules are CIATS-owned starting values. They will be replaced by data during paper trading and live trading.

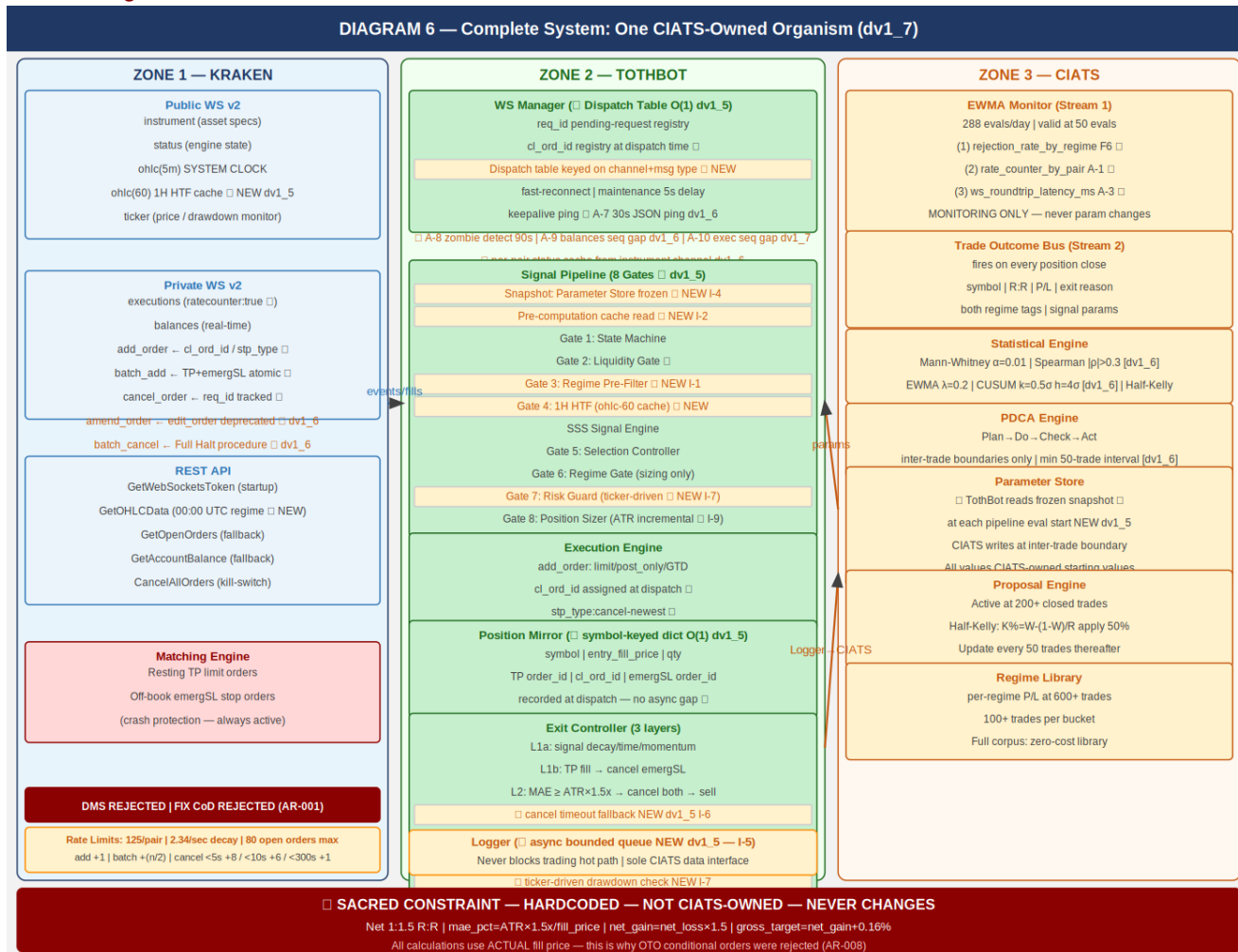


Diagram 6: Complete system. Three zones, one organism. A-10 executions seq gap detection new in dv1_7.

8. Locked Architectural Decisions

ID	Decision	Specification
AR-001	No Dead Man Switch	Crash protection is resting TP + emergSL on matching engine. DMS and FIX cancel-on-disconnect both REJECT
AR-002	Position Mirror	Symbol-keyed dict, O(1) lookup. (NEW dv1_5 — I-3)
AR-003	Restart reconciliation	PREFERRED: snap_orders:true. FALLBACK: REST.
AR-004	Two API key pairs	DATA + TRADE. Nonce window 10,000ms.
AR-005	WS v2 only	All future Kraken enhancements in v2 only.
AR-006	Entry order	limit/post_only/GTD/cl_ord_id/stp_type:cancel-newest.
AR-007	TP placement	Resting limit via batch_add on fill. Actual fill price for R:R.
AR-008	Emergency SL	Stop-market via batch_add atomic with TP. ATR(14) × 3.0x. OTO REJECTED: uses limit price, violates R:R, Pos
AR-009	Short deferred	Long Module only for V1.
AR-010	24/7/365	CIATS adds stand-down via PDCA if data supports.
AR-011	Net 1:1.5 R:R	SACRED — HARDCODED — NOT CIATS-OWNED. Never changes.
AR-012	Fee baseline	Maker 0.16% / Taker 0.26%.
AR-013	Kraken Pro Intermediate tier	Confirmed.
AR-014	Logger async bounded queue	10,000 entries. Never blocks hot path. (dv1_5 I-5)
AR-015	WS dispatch table	O(1) routing on channel+message type. (dv1_5 I-8)
AR-016	ATR(14) incremental	O(1) per update. (dv1_5 I-9)
AR-017	Daily regime	REST GetOHLCData at 00:00 UTC. CRITICAL: exclude response[-1]. (dv1_5 FLAG 2)
AR-018 ★ dv1_6	amend_order PRIMARY	edit_order DEPRECATED. amend_order is the only amendment method.
AR-019 ★ dv1_6	batch_cancel	Used for Full Halt multi-order cancellation.
AR-020 ★ dv1_6	Per-pair status cache	From instrument channel. Gates 1/2 pre-filter.
AR-021 ★ dv1_7	Executions seq gap detection	Any gap → REST GetOpenOrders reconciliation. Missed executions = Position Mirror desync = loss risk.

9. Complete CIATS Parameter Registry

ALL PARAMETERS ARE CIATS-OWNED STARTING VALUES.

Parameter	Starting Value	Notes
tradeable_pct	50% of portfolio	CIATS-owned
per_trade_pct	5% → Half-Kelly at 200 trades	Update every 50 trades thereafter
max_concurrent	20 positions	CIATS-owned
mae_mult (ATR x)	1.5x ATR(14)	CIATS-owned
emergency_sl_mult	3.0x ATR(14)	CIATS-owned
entry_timeout_sec	30-60s GTD	CIATS-owned
full_halt_drawdown	10% of portfolio	CIATS-owned
session_pause_drawdown	5% of portfolio	CIATS-owned
adx_threshold	25	CIATS-owned
atr_percentile_threshold	67th	CIATS-owned
htf_ema_periods	20/50 daily	CIATS-owned
min_volume_usd_daily	\$500k USD	CIATS-owned
1:1.5 R:R net	HARDCODED	★★★ SACRED — NOT CIATS-OWNED — NEVER CHANGES ★★★

10. Six FP/DP Validated System Improvements

Derived from 80+ sources across 3 independent research passes plus 2 deep validation sessions (144+ sources) plus two 10-search Kraken validation sessions [dv1_6, dv1_7] (239+ sources total). 5+ sources per parameter.

Finding 1: 1H HTF Confirmation Gate [TothBot]	Gap: No layer between daily regime and 5-min entry. Evidence: 15-20% higher win rates with daily+hourly filter (5 sources 2024-2025). Change: 1H EMA(20) > EMA(50) required for TRENDING_POSITIVE. Cache from ohlc(60). Effect: Removes false signals during 1H pullbacks. Zero API cost.
Finding 2: ATR Dynamic Stop Distances [TothBot/Exit]	Gap: Fixed 2% stop ignores real-time market volatility. Evidence: +15% performance, -22% max drawdown vs fixed stops (5 sources). Change: MAE ATR(14) x 1.5x. Emergency SL ATR(14) x 3.0x. Incremental (I-9). Effect: Stop adapts to conditions. R:R formula preserved.
Finding 3: Regime Gate [TothBot]	Gap: Long momentum system trading in absent-edge conditions. Evidence: Accuracy 65%→45-50% in NON_DIRECTIONAL. Break-even 40%. Change: NON_DIR+ELEV and TRENDING_NEG: no entry (Gate 3). NON_DIR+NORM: 50% (Gate 6). Effect: Stops trading when documented edge is absent. Pure loss minimization.
Finding 4: Minimum Liquidity Gate [TothBot]	Gap: No minimum 24h volume on asset selection. Evidence: \$1M+/day minimum for clean execution (5 sources 2024-2025). Change: min_24h_volume_usd \$500k starting. Checked daily via ticker. Effect: Protects fee formula. Ensures clean post_only fills and SL execution.
Finding 5: CIATS Half-Kelly Pathway [CIATS]	Gap: Fixed 5% sizing had no defined data-driven transition. Evidence: Half-Kelly: 75% optimal growth at 50% less drawdown (5 sources). Change: At 200 trades: $K\% = W - (1-W)/R$, apply 50%. Update every 50 trades. Effect: Data-driven sizing replaces arbitrary fixed fraction.
Finding 6: post_only Rejection Monitoring [CIATS]	Gap: Rejection events logged but unused by CIATS EWMA. Evidence: Rejection rate = direct signal of entry placement quality (5 sources). Change: CIATS EWMA adds rejection_rate_by_regime stream. Effect: Free diagnostic signal. Entry optimization pathway activated.

11. Complete Trade Lifecycle — Narrative Reference

11.1 Normal Trade (Entry to TP or Exit Controller)

- OHLC 5-min candle close fires on public WS v2
- **Parameter Store snapshot frozen** — all CIATS params captured (NEW dv1_5 — I-4)
- Pre-computation cache read — regime, 1H EMA, ATR(14) incremental, 24h liquidity (I-2)
- **★ Per-pair instrument status check — skip reduce_only/work_in_progress/delisted [dv1_6]**
- **Gate 1: State Machine** — online or post_only: continue; else halt and wait
- **Gate 2: Liquidity Gate** — 24h volume >= \$500k: continue; else skip
- **Gate 3: Regime Pre-Filter** — TRENDING_NEG or NON_DIR+ELEV: STOP; else continue (I-1)
- **Gate 4: 1H HTF Gate** — 1H EMA(20) > EMA(50) for TRENDING_POS: continue; else skip
- Signal Engine: SSS score computed on 5-min candle
- **Gate 5: Selection Controller** — 6-gate quality evaluation: pass or reject
- **Gate 6: Regime Gate** — TRENDING_POS: proceed; NON_DIR+NORM: 50% size
- **Gate 7: Risk Guard — drawdown/concentration/exposure checks: pass or halt**
- **Gate 8: Position Sizer** — MAE=ATR × 1.5x, emergSL=ATR × 3.0x, TP=R:R formula
- Execution: WS add_order (limit/post_only/GTD 30-60s/cl_ord_id/stp_type:cancel-newest)
- If expired/rejected: log reason code → CIATS EWMA monitors rejection rate. Done.
- If filled: WS batch_add → TP (resting limit) + emergSL (stop-market) atomic
- Position Mirror records: symbol (key), entry_fill_price, qty, TP id, cl_ord_id, emergSL id
- Exit Controller monitors via ticker (continuous drawdown check — I-7)
- Logger records full entry via async queue. CIATS Trade Outcome Bus ready.
- TP fills: EC cancels emergSL. Or EC fires Layer 1/2 exit.
- Logger records full exit. CIATS Trade Outcome Bus fires with complete trade record.

11.2 Layer 2 MAE Exit (Adverse Price)

- Exit Controller detects: current_price - entry >= ATR(14) × 1.5x
- WS cancel_order: cancel TP resting order
- **WS cancel_order: cancel emergSL stop order (SEQUENCE CRITICAL: both cancels before sell)**
- **★ CANCEL TIMEOUT FALLBACK (NEW dv1_5 — I-6): req_id registry tracks each cancel ACK. On timeout: check executions channel state. Confirmed cancelled: proceed. State unknown: retry once. Second timeout: ALERT and hold position. NEVER issue market sell with ambiguous order state.**
- WS add_order: market sell (taker 0.26%)
- **★ MPP NOTE (C-1): Market sell subject to MPP. EC MUST handle MPP rejection and retry.**
- Logger: exit reason = MAE_THRESHOLD_BREACH
- CIATS Trade Outcome Bus fires

12. Open Items — Require Specification Documents

OI	Description	Blocks
OI-001	Signal generation (SSS) — FP/DP derivation required	0811001, 0811002
OI-002	Asset selection universe — Kraken USDT/USDC pairs	0711003
OI-003	Exit Controller full specification — MPP handling, cancel timeout, ticker drawdown, batch_cancel	0811003
OI-004	Entry limit placement — evaluate bbo trigger, L2 book, L3 book	0811004
OI-005	Selection Controller — 6-gate spec, must include per-pair status check	0811002
OI-006	CIATS specification — all dv1_6 statistical thresholds	0211001
OI-007	AA=04 Exchange Interface formal document	0400000
OI-008	All AA=10 coding specs. WD documents active: 1011001, 1011002, 1011010, 1011012	Phase 8

13. Revision History

Version	Date	Changes
dv1_0	03-23-2026	Initial document.
dv1_1	03-23-2026	Added all six system flow diagrams.

dv1_2	03-24-2026	Six changes: C-1 emergSL terminology, C-2 rate limits, C-3 OHLC stall, A-1 ratecounter, A-2 req_id, A-3 latency logging.
dv1_3	03-24-2026	Rebuilt as correct .docx with all six diagrams embedded.
dv1_4	03-24-2026	Seven changes: C-1 MPP, A-1 AR-001 expansion, A-2 cl_ord_id, A-3 snap_orders, A-4 stp_type, A-5 reconnect distinction
dv1_5	03-24-2026	Eleven changes: FLAG1 ohlc(60), FLAG2 REST GetOHLCData, I-1 Gate 3, I-2 cache, I-3 Position Mirror O(1), I-4 Param
dv1_6	03-24-2026	Six changes from 10-search FP/DP validation: (1) amend_order primary, edit_order deprecated, batch_cancel, L3 URL co
dv1_7	03-25-2026	★ One change from 10-search FP/DP organism validation (TB00009): A-10: Executions channel sequence gap detection.